ST5209/X Assignment 5

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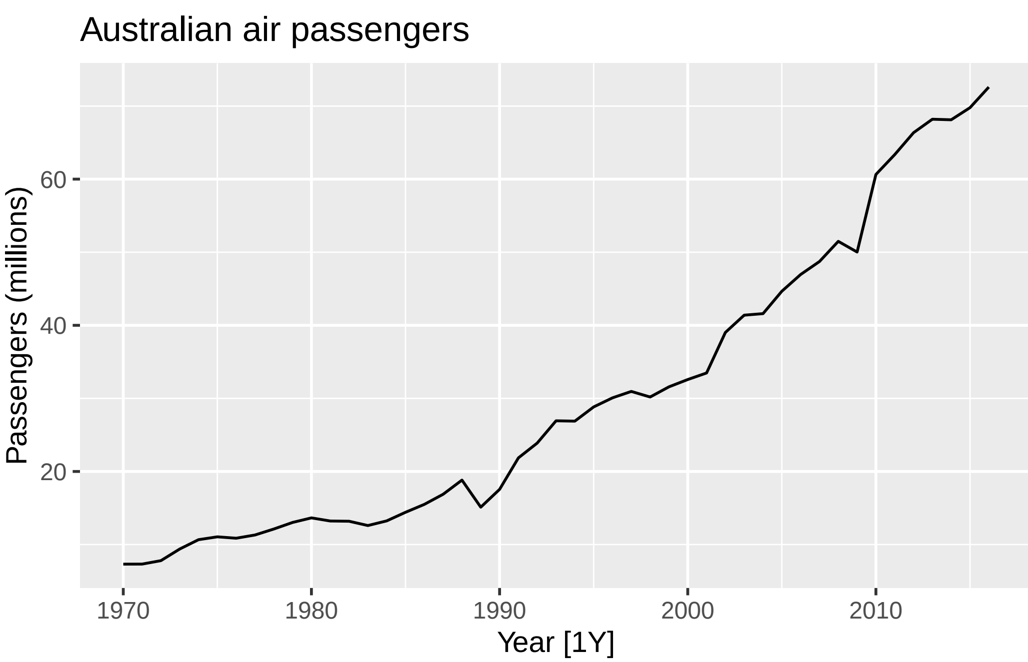
Submission

1. Render the document to get a .pdf printout.
2. Submit both the .qmd and .pdf files to Canvas.

1. Modeling with ARIMA (Q9.7 in Hyndman & Athanasopoulos)

```
aus_airpassengers |> autoplot(Passengers) +
  labs(title = "Australian air passengers", y = "Passengers (millions)")
```



a. **Fit an ARIMA model with `ARIMA()`.**

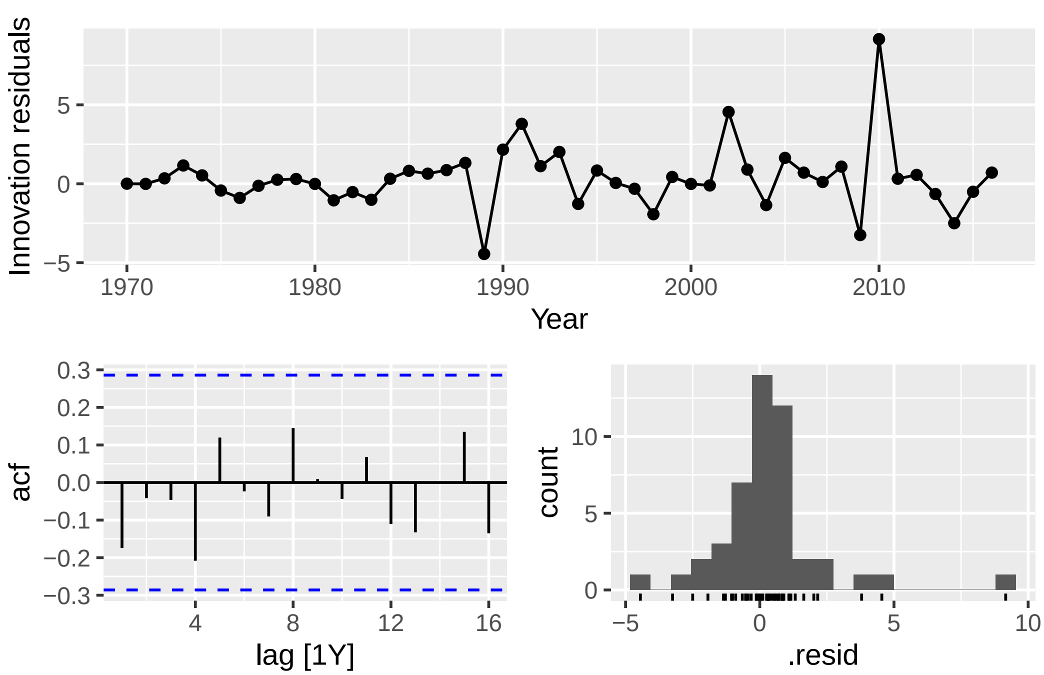
```r
fit_auto <- aus_airpassengers |>
  model(arima = ARIMA(Passengers ~ 0 + pdq(0, 2, 1)))
report(fit_auto)
```

```
Series: Passengers
Model: ARIMA(0,2,1)

Coefficients:
          ma1
      -0.8963
s.e.   0.0594

sigma^2 estimated as 4.308:  log likelihood=-97.02
AIC=198.04   AICc=198.32   BIC=201.65
```

The selected model is **ARIMA(0, 2, 1)** (no constant). The estimated MA(1) coefficient is $\hat{\theta}_1 = -0.8963$ (s.e. 0.0594), $\hat{\sigma}^2 = 4.308$, with log-likelihood -97.02, AIC $= 198.04$, AICc $= 198.32$, BIC $= 201.65$.

Residual diagnostics.

```r
fit_auto |> gg_tsresiduals()
```



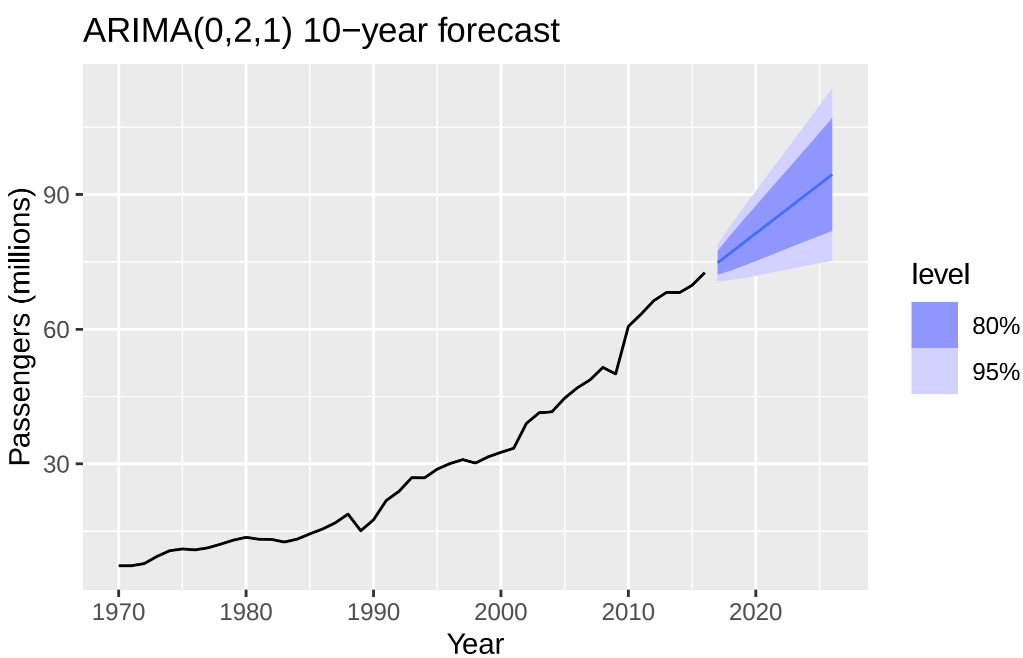
```r
fit_auto |> augment() |> features(.innov, ljung_box, lag = 10, dof = 1)
```

```
# A tibble: 1 x 3
  .model lb_stat lb_pvalue
  <chr>    <dbl>     <dbl>
1 arima     6.70     0.669
```

The residual ACF spikes all sit inside the significance bands (largest $|\hat{\rho}| \approx 0.21$ at lag 4, within $\pm 2/\sqrt{47} \approx \pm 0.29$), the residual histogram looks roughly symmetric, and the Ljung–Box test at lag 10 (with `dof = 1`) gives $Q^* = 6.70$ and $p = 0.669$. We do not reject the white-noise null, so the residuals are consistent with white noise.

Forecasts for the next 10 years.

```r
fit_auto |>
  forecast(h = 10) |>
  autoplot(aus_airpassengers) +
  labs(title = "ARIMA(0,2,1) 10-year forecast", y = "Passengers (millions)")
```



The point forecasts from 2017 to 2026 are approximately $74.8, 77.0, 79.2, 81.3, 83.5, 85.7, 87.9, 90.1, 92.3, 94.5$ (in millions) — a nearly linear extrapolation with slope ≈ 2.2 per year.

b. **Backshift operator form.**

For ARIMA(0, 2, 1), the model is $(1-B)^2X_t = (1+\theta_1 B)W_t$. Substituting $\hat{\theta}_1 = -0.8963$,

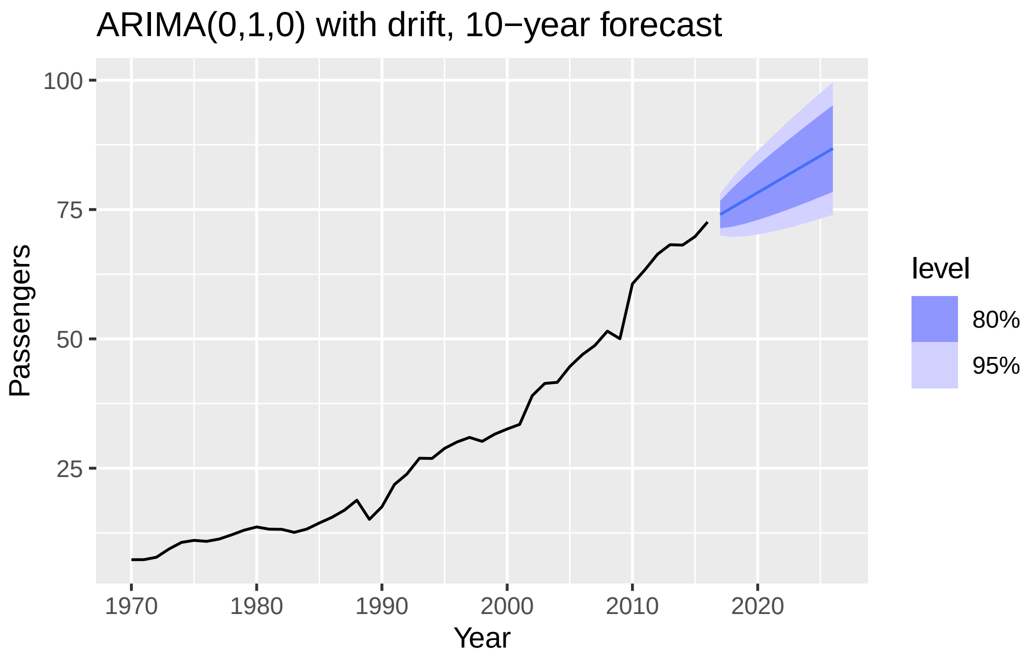
$$(1-B)^2X_t = (1-0.8963\,B)\,W_t, \qquad W_t \sim WN(0, 4.308).$$

c. **ARIMA(0, 1, 0) with drift.**

```
fit_c <- aus_airpassengers |>
  model(arima = ARIMA(Passengers ~ 1 + pdq(0, 1, 0)))
report(fit_c)
```

```
Series: Passengers
Model: ARIMA(0,1,0) w/ drift

Coefficients:
      constant
        1.4191
s.e.    0.3014

sigma^2 estimated as 4.271:  log likelihood=-98.16
AIC=200.31   AICc=200.59   BIC=203.97
```

```
fit_c |>
  forecast(h = 10) |>
  autoplot(aus_airpassengers) +
  labs(title = "ARIMA(0,1,0) with drift, 10-year forecast")
```



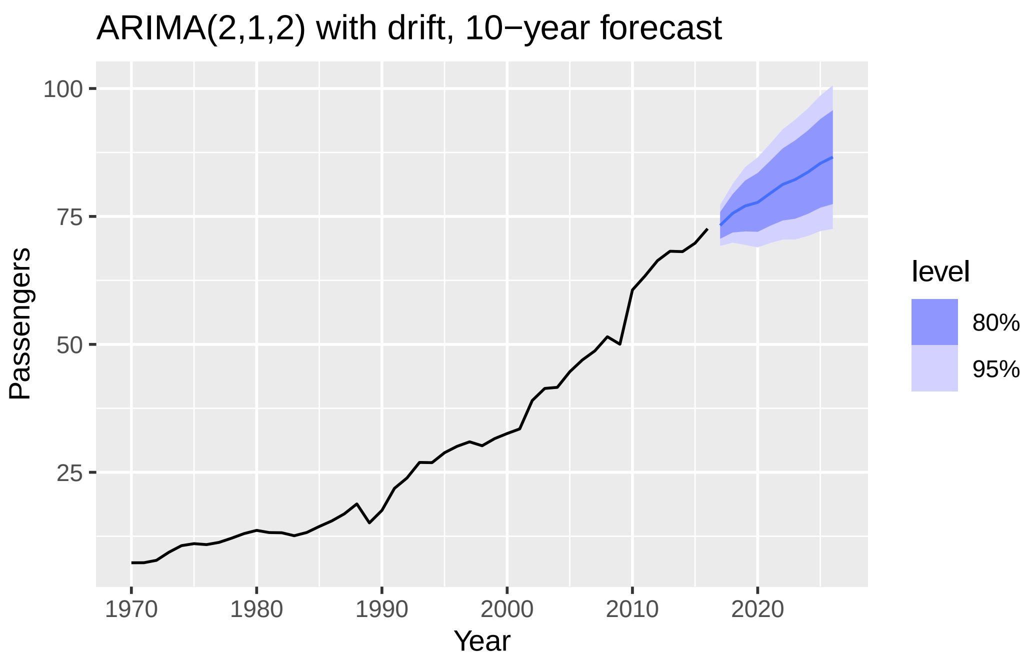
This is a random walk with drift $\hat{c} = 1.4191$ per year (s.e. 0.3014, $\hat{\sigma}^2 = 4.271$, AICc = 200.59). The point forecasts are $74.0, 75.4, 76.9, 78.3, 79.7, 81.1, 82.5, 84.0, 85.4, 86.8$ — a straight line with slope $\hat{c} = 1.4191$. Compared with part (a), both give straight-line forecasts, but the slope here (1.42) is smaller than that from ARIMA(0,2,1) (≈ 2.2). Formally, ARIMA(0,2,1) implies that the first differences follow an MA(1), so the forecast extrapolates the *recent* change in slope rather than a fixed drift, which better captures the steepening growth near the end of the series.

d. **ARIMA(2, 1, 2) with drift.**

```r
fit_d <- aus_airpassengers |>
  model(arima = ARIMA(Passengers ~ 1 + pdq(2, 1, 2)))
report(fit_d)
```

```
Series: Passengers
Model: ARIMA(2,1,2) w/ drift

Coefficients:
          ar1      ar2     ma1     ma2  constant
      -0.5518  -0.7327  0.5895  1.0000    3.2216
s.e.   0.1684   0.1224  0.0916  0.1008    0.7224

sigma^2 estimated as 4.031:  log likelihood=-96.23
AIC=204.46   AICc=206.61   BIC=215.43
```

```r
fit_d |>
  forecast(h = 10) |>
  autoplot(aus_airpassengers) +
  labs(title = "ARIMA(2,1,2) with drift, 10-year forecast")
```



The fit gives $\hat{\phi}_1 = -0.552, \hat{\phi}_2 = -0.733, \hat{\theta}_1 = 0.590, \hat{\theta}_2 = 1.000$, constant 3.222, $\hat{\sigma}^2 = 4.031$, AICc = 206.61 (worse than part (a)'s 198.32). Forecasts 73.2, 75.6, 77.1, 77.7, 79.5, 81.3, 82.2, 83.6, 85.4, 86.6 show a mild wave pattern (complex AR roots) superimposed on an upward trend, tracking slightly below the ARIMA(0,2,1) forecasts in the long run.

Removing the constant.

```r
fit_d2 <- tryCatch(
  aus_airpassengers |>
    model(arima = ARIMA(Passengers ~ 0 + pdq(2, 1, 2))),
  error = function(e) e
)
fit_d2
```

```
# A mable: 1 x 1
          arima
        <model>
1 <NULL model>
```

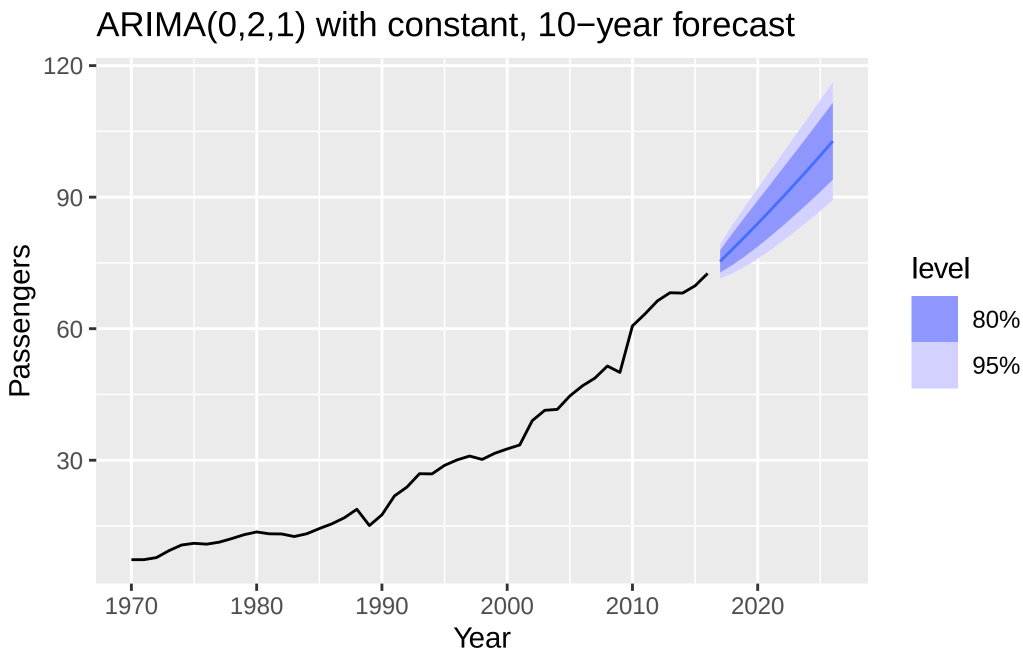
Without the drift term, `ARIMA()` returns a `NULL model` and reports *"non-stationary AR part from CSS"* — the conditional-sum-of-squares optimiser cannot find a stationary AR configuration for this specification, so estimation fails.

e. **ARIMA(0, 2, 1) with a constant.**

```
fit_e <- aus_airpassengers |>
  model(arima = ARIMA(Passengers ~ 1 + pdq(0, 2, 1)))
report(fit_e)
```

```
Series: Passengers
Model: ARIMA(0,2,1) w/ poly

Coefficients:
          ma1  constant
      -1.0000    0.0571
s.e.   0.0585    0.0213

sigma^2 estimated as 3.855:  log likelihood=-95.1
AIC=196.21   AICc=196.79   BIC=201.63
```

```
fit_e |>
  forecast(h = 10) |>
  autoplot(aus_airpassengers) +
  labs(title = "ARIMA(0,2,1) with constant, 10-year forecast")
```



`ARIMA()` warns that the specification induces a polynomial trend and labels the model `ARIMA(0,2,1) w/ poly`. With $d = 2$, a non-zero constant c means the second difference has mean c, so X_t carries a quadratic term $\frac{c}{2}t^2$. The forecasts $75.4, 78.2, 81.1, 84.0, 87.0, 90.0, 93.1, 96.3, 99.5, 103.0$ have increasing year-on-year increments (2.8, 2.9, 2.9, 3.0, 3.0, 3.1, 3.1, 3.2, 3.3), i.e. curve upwards. Such quadratic extrapolation is unrealistic for air-passenger demand; the model in (a) is preferred.

2. Seasonal ARIMA

a. **Load the data.**

```r
diabetes <- read_rds("../_data/cleaned/diabetes.rds")
head(diabetes)
```

```
# A tsibble: 6 x 3 [1M]
     Month  TotalC  Cost
     <mth>   <dbl> <dbl>
1 1991 Jul 3526591  3.53
2 1991 Aug 3180891  3.18
3 1991 Sep 3252221  3.25
4 1991 Oct 3611003  3.61
5 1991 Nov 3565869  3.57
6 1991 Dec 4306371  4.31
```

b. **Build `Y`: log + seasonal difference.**

```
dbt <- diabetes |> mutate(Y = difference(log(Cost), lag = 12))
```

c. **KPSS test on `Y`.**

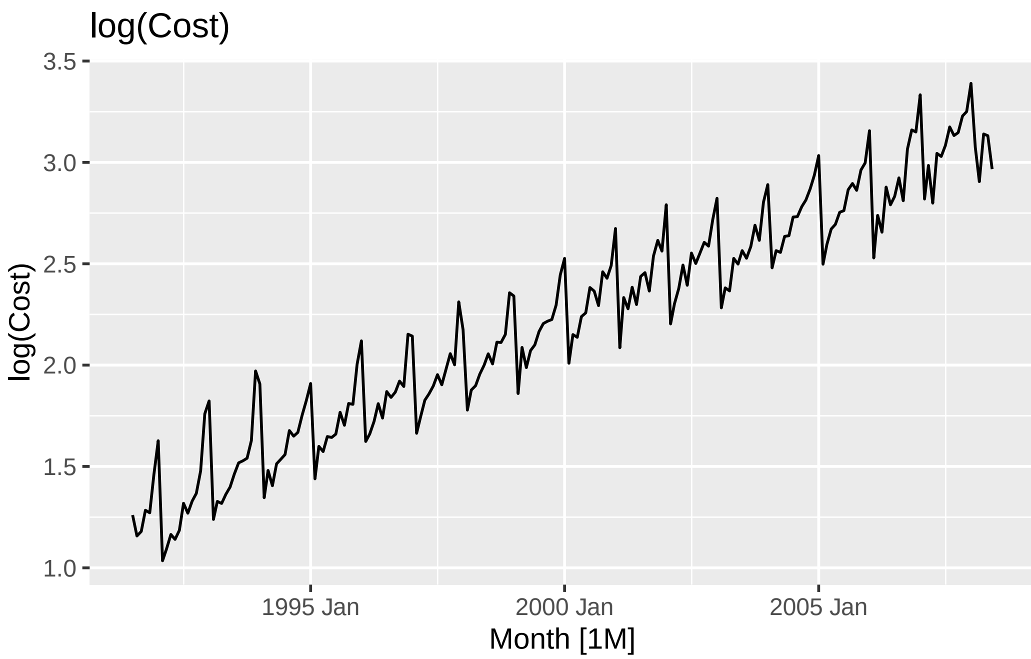
```
dbt |> features(Y, unitroot_kpss)
```

```
# A tibble: 1 x 2
  kpss_stat kpss_pvalue
      <dbl>       <dbl>
1     0.209         0.1
```

The KPSS statistic is 0.209 with reported p-value 0.1 (the feasts/urca implementation caps p-values above 0.1 at 0.1). Since $p \geq 0.1 > 0.05$, we **fail to reject** the null of stationarity, so `Y` is consistent with a (level-)stationary series.

d. **Time plot of `log(Cost)`.**

```
dbt |> autoplot(log(Cost)) +
  labs(title = "log(Cost)")
```



`log(Cost)` shows a clear upward trend plus a strong annual seasonal pattern. The trend disappears from `Y` for two reasons at once:

- If $\log(\text{Cost}_t) \approx \alpha + \beta t + s_t + u_t$ with period-12 seasonal $s_t = s_{t-12}$ and linear trend, then $Y_t = 12\beta + (u_t - u_{t-12})$, i.e. the trend collapses to the constant 12β and the seasonal component cancels out.
- More generally, a seasonal difference removes any slowly-varying component whose value at t is close to its value at $t-12$, which is exactly the behaviour of a linear (or locally linear) trend. What is left is short-horizon variation, which KPSS does not detect as non-stationary.

e. **Fit an ARIMA model to `log(Cost)`.**

```r
fit2 <- dbt |> model(arima = ARIMA(log(Cost)))
report(fit2)
```

```
Series: Cost
Model: ARIMA(2,0,2)(1,1,1)[12] w/ drift
Transformation: log(Cost)

Coefficients:
         ar1     ar2      ma1     ma2    sar1     sma1  constant
      0.7211  0.1081  -0.6809  0.1198  0.2236  -0.8464    0.0151
s.e.  0.2364  0.2064   0.2316  0.1752  0.1112   0.0835    0.0004

sigma^2 estimated as 0.003595:  log likelihood=266.01
AIC=-516.02   AICc=-515.24   BIC=-489.96
```

The fitted model is $\mathbf{ARIMA(2,0,2)(1,1,1)_{12}}$ **with drift**, with coefficients $\hat{\phi}_1 = 0.721, \hat{\phi}_2 = 0.108, \hat{\theta}_1 = -0.681, \hat{\theta}_2 = 0.120, \hat{\Phi}_1 = 0.224, \hat{\Theta}_1 = -0.846$, drift $= 0.0151$, $\hat{\sigma}^2 = 0.003595$, log-likelihood $= 266.01$, AIC $= -516.02$, AICc $= -515.24$, BIC $= -489.96$.

f. **Number of fitted parameters.**

```r
tidy(fit2)
```

```
# A tibble: 7 x 6
  .model term     estimate std.error statistic  p.value
  <chr>  <chr>       <dbl>     <dbl>     <dbl>    <dbl>
1 arima  ar1        0.721   0.236        3.05  2.61e- 3
2 arima  ar2        0.108   0.206        0.524 6.01e- 1
3 arima  ma1       -0.681   0.232       -2.94  3.69e- 3
4 arima  ma2        0.120   0.175        0.684 4.95e- 1
5 arima  sar1       0.224   0.111        2.01  4.56e- 2
6 arima  sma1      -0.846   0.0835     -10.1   1.32e-19
7 arima  constant   0.0151  0.000426    35.4   5.43e-86
```

The `tidy()` output lists **7 fitted coefficients**: `ar1`, `ar2`, `ma1`, `ma2`, `sar1`, `sma1`, `constant`. If we also count the innovation variance σ^2 as a fitted parameter (which R's AIC formula does), there are $k = 8$ parameters — this matches the implied penalty because $\text{AIC} - (-2 \log L) = -516.02 + 2(266.01) = 16 = 2 \times 8$.

3. Model selection

a. **Null hypothesis and assumptions of the ADF test.**

The Augmented Dickey–Fuller test fits a regression of ΔX_t on X_{t-1} and lagged differences $\Delta X_{t-1}, \ldots, \Delta X_{t-p}$, optionally with a constant and linear time trend. The null hypothesis is H_0 : the series has a **unit root** (is non-stationary, specifically $I(1)$), against H_1 : the series is stationary (possibly around a trend). The main assumptions are:

- The true data-generating process is a linear autoregression with i.i.d. (or at least white-noise, mean-zero, finite-variance) innovations;
- The number of lagged differences p is chosen large enough to make the residuals uncorrelated but not so large that power is lost;
- Any deterministic component (constant / trend) included in the test regression matches the deterministic structure of the data under H_1.

b. **Can ADF and KPSS both have large p-values?**

Yes. The two tests have *opposite* null hypotheses — ADF tests H_0 : unit root vs KPSS tests H_0 : stationarity — so "large p-value" means *failing to reject* their respective nulls:

- Large ADF p-value $\Rightarrow$ we cannot rule out a unit root.
- Large KPSS p-value $\Rightarrow$ we cannot rule out stationarity.

Both tests can have low power in short or borderline samples (e.g. near-unit-root processes, $\phi = 0.95$), so it is entirely possible for neither test to reject. The correct interpretation then is simply that the data are not informative enough to distinguish stationarity from a unit root, and additional information (differencing + re-testing, plots, domain knowledge) should be used.

c. **AIC and AICc penalties for the Q2 model.**

With $k = 8$ fitted parameters (7 coefficients + σ^2) and $n = 204$ observations:

```r
n <- nrow(diabetes); k <- 8
aic_penalty  <- 2 * k
aicc_penalty <- 2 * k + 2 * k * (k + 1) / (n - k - 1)
c(AIC_penalty = aic_penalty, AICc_penalty = aicc_penalty)
```

```
 AIC_penalty AICc_penalty
    16.00000     16.73846
```

- AIC penalty $= 2k = \mathbf{16}$.
- AICc penalty $= 2k + \dfrac{2k(k+1)}{n-k-1} = 16 + \dfrac{2 \cdot 8 \cdot 9}{204-8-1} = 16 + \dfrac{144}{195} \approx \mathbf{16.738}$.

Sanity check: $\text{AIC} - (-2 \log L) = -516.02 + 532.02 = 16\checkmark$, $\text{AICc} - \text{AIC} \approx 0.787 \approx 144/195\checkmark$.

d. **Fit an ETS model and compare.**

```r
fit_both <- diabetes |>
  model(
    arima = ARIMA(log(Cost)),
    ets   = ETS(Cost)
  )
glance(fit_both) |> select(.model, log_lik, AIC, AICc, BIC)
```

```
# A tibble: 2 x 5
  .model log_lik   AIC  AICc   BIC
  <chr>    <dbl> <dbl> <dbl> <dbl>
1 arima     266. -516. -515. -490.
2 ets      -417.  868.  871.  924.
```

The ETS selector chooses **ETS(M, A, M)** (multiplicative error, additive trend, multiplicative seasonality). The comparison table gives

Model	log L	AICc
ARIMA(2,0,2)$(1,1,1)_{12}$ w/ drift on log(Cost)	+266.01	−515.24
ETS(M, A, M) on Cost	−416.97	871.23

e. **Can we compare them via log-likelihood or AICc?**

No, not directly. The ARIMA was fit on log(Cost), so its likelihood is on the log scale, while ETS(M,A,M) was fit on `Cost`. The change of variables introduces a Jacobian that shifts the ARIMA log-likelihood by $-\sum_t \log(\text{Cost}_t)$, which is a data-dependent constant unrelated to model fit. To compare the two, either fit them on the same scale, adjust the ARIMA log-likelihood by $-\sum_t \log(\text{Cost}_t)$, or compare out-of-sample forecast accuracy via time series cross-validation (scale-free).

4. Dynamic regression (adapted from Q10.4 in Hyndman & Athanasopoulos)

a. **Inflation adjustment.**

```
acc <- aus_accommodation |>
  mutate(AdjTakings = Takings / CPI * 100)
```

b. **Fit a dynamic regression model for each state.**

```
knot <- yearquarter("2008 Q1")
fits <- acc |>
  model(dr = ARIMA(AdjTakings ~ season() + trend(knots = knot)))
fits
```

```
# A mable: 8 x 2
# Key:     State [8]
  State                                                            dr
  <chr>                                                       <model>
1 Australian Capital Territory           <LM w/ ARIMA(1,0,0) errors>
2 New South Wales              <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
3 Northern Territory           <LM w/ ARIMA(0,0,1)(1,0,0)[4] errors>
4 Queensland                   <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
5 South Australia              <LM w/ ARIMA(1,0,0)(1,0,0)[4] errors>
6 Tasmania                     <LM w/ ARIMA(0,0,1)(1,0,0)[4] errors>
7 Victoria                     <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
8 Western Australia                      <LM w/ ARIMA(1,0,0) errors>
```

Each row shows which ARIMA error structure was chosen by `ARIMA()` state-by-state.

c. **Victoria.**

```
vic <- fits |> filter(State == "Victoria")
report(vic)
```

```
Series: AdjTakings
Model: LM w/ ARIMA(1,0,0)(0,0,1)[4] errors

Coefficients:
         ar1    sma1  season()year2  season()year3  season()year4
      0.6596  0.3756       -57.8760       -36.0249       -12.9360
s.e.  0.0866  0.1242         3.9693         4.6151         4.0447
```

```
      trend(knots = knot)trend  trend(knots = knot)trend_41  intercept
                        3.1995                       0.3892   264.4398
s.e.                    0.4871                       0.9790    13.3769

sigma^2 estimated as 170.8:  log likelihood=-291.63
AIC=601.26   AICc=604.07   BIC=622
```

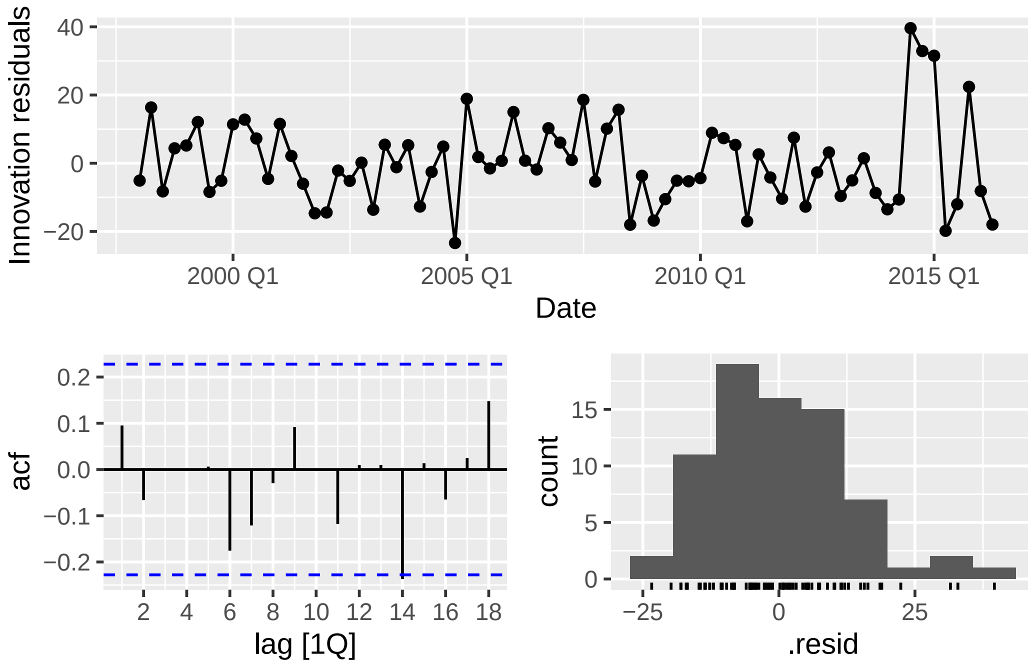
For Victoria, the fitted model is a linear regression with $\mathbf{ARIMA(1,0,0)(0,0,1)}_4$ errors. The regression part has quarterly dummies, a pre-2008 slope, and a kink at 2008 Q1:

- Intercept $= 264.44$;
- Seasonal dummies (Q1 is the reference level): year2 $= -57.88$, year3 $= -36.02$, year4 $= -12.94$ — all highly significant — so Q1 has the highest takings, followed by Q4, Q3, Q2;
- Pre-knot linear slope $= 3.20$ (significant, $p \approx 6 \times 10^{-9}$); the post-knot *additional* slope is 0.389 and is **not significant** ($p = 0.69$), so the trend essentially continues at the same pre-2008 rate;
- AR(1) $= 0.660$, SMA(1) $= 0.376$ (both significant, $p < 0.005$).

Is seasonality constant? The seasonal dummies impose a *deterministic, time-invariant* seasonal pattern. The significant seasonal MA(1) term ($\hat{\Theta}_1 = 0.376$, $p = 0.003$) indicates that **residual seasonal dependence remains after the fixed dummies** — the dummies alone do not fully capture the seasonal structure, and the SMA(1) component absorbs the leftover stochastic seasonal correlation.

d. **Residual diagnostics for Victoria.**

```
vic |> gg_tsresiduals()
```




```
vic |> augment() |> features(.innov, ljung_box, lag = 8, dof = 0)
```

```
# A tibble: 1 x 4
  State    .model lb_stat lb_pvalue
  <chr>    <chr>    <dbl>     <dbl>
1 Victoria dr        4.90     0.769
```

The innovation-ACF plot shows no spike outside the significance bands and the Ljung–Box test at lag 8 gives $Q^* = 4.90$ with $p = 0.769$ — far above any reasonable significance level — so the residuals are consistent with white noise.

e. **Forecast CPI, then forecast Victoria takings to the end of 2017.**

The data ends in 2016 Q2, so we need $h = 6$ quarters (2016 Q3 through 2017 Q4).

```
# Forecast CPI separately with its own ARIMA model
cpi_ts <- acc |>
  filter(State == "Victoria") |>
  select(Date, CPI) |>
  as_tsibble(index = Date)
fit_cpi <- cpi_ts |> model(cpi_arima = ARIMA(CPI))
report(fit_cpi)
```

```
Series: CPI
Model: ARIMA(0,1,0) w/ drift

Coefficients:
      constant
        0.5699
s.e.    0.0520

sigma^2 estimated as 0.2003:  log likelihood=-44.37
AIC=92.74   AICc=92.91   BIC=97.32
```

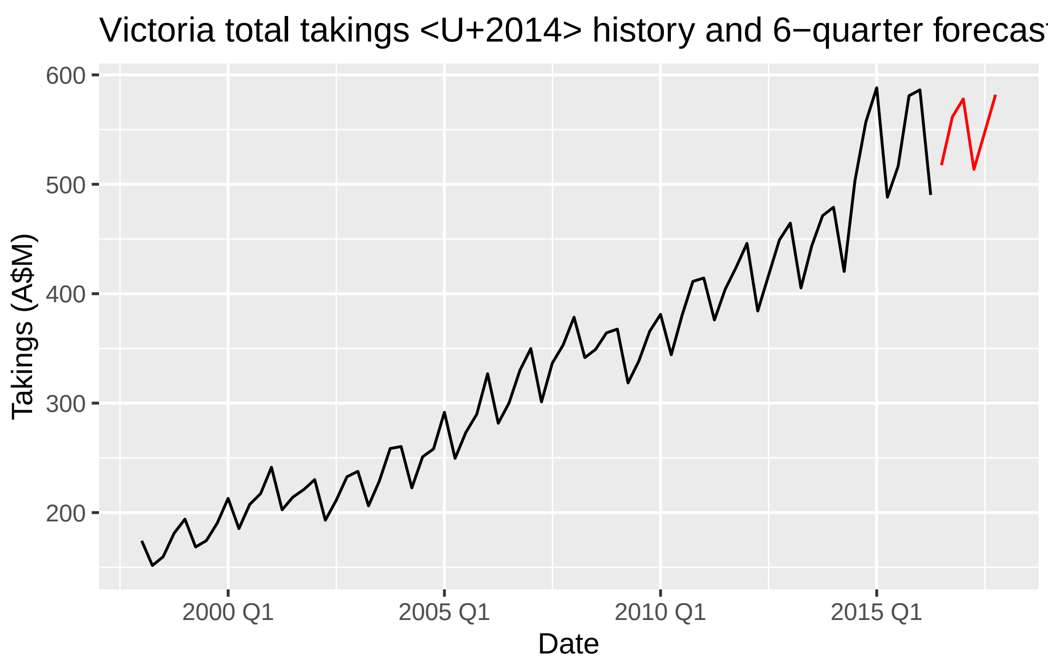
```r
fcst_cpi <- fit_cpi |> forecast(h = 6)

# Supply forecasted CPI as new_data to the Victoria dynamic regression
new_data_vic <- new_data(acc |> filter(State == "Victoria"), 6) |>
  mutate(CPI = fcst_cpi$.mean)

fcst_adj <- vic |> forecast(new_data = new_data_vic)

takings_fcst <- as_tibble(fcst_adj) |>
  select(Date, AdjTakings_hat = .mean) |>
  mutate(CPI = new_data_vic$CPI,
         Takings_hat = AdjTakings_hat * CPI / 100)
takings_fcst
```

```
# A tibble: 6 x 4
     Date AdjTakings_hat   CPI Takings_hat
    <qtr>          <dbl> <dbl>       <dbl>
1 2016 Q3           474.  109.        517.
2 2016 Q4           512.  110.        562.
3 2017 Q1           524.  110.        578.
4 2017 Q2           463.  111.        514.
5 2017 Q3           491.  111.        547.
6 2017 Q4           519.  112.        582.
```

The CPI model chosen is **ARIMA(0,1,0) with drift** (random walk with drift $\hat{c} = 0.570$ per quarter). Plugging the point forecasts of CPI into the Victoria model and re-inflating the adjusted takings by $\widehat{\text{Takings}} = \widehat{\text{AdjTakings}} \cdot \widehat{\text{CPI}}/100$ gives:

Quarter	$\widehat{\text{AdjTakings}}$	$\widehat{\text{CPI}}$	$\widehat{\text{Takings}}$ (A$M)
2016 Q3	473.99	109.17	**517.45**
2016 Q4	511.67	109.74	**561.50**
2017 Q1	523.98	110.31	**578.01**
2017 Q2	463.23	110.88	**513.62**
2017 Q3	491.15	111.45	**547.39**
2017 Q4	519.47	112.02	**581.90**

```
acc |>
  filter(State == "Victoria") |>
  ggplot(aes(x = Date, y = Takings)) +
  geom_line() +
  geom_line(data = takings_fcst, aes(y = Takings_hat), colour = "red") +
  labs(title = "Victoria total takings — history and 6-quarter forecast",
       y = "Takings (A$M)")
```



The forecast preserves the strong Q1-peak / Q2-trough seasonal pattern and continues the pre-2008 linear rise.

f. **Sources of uncertainty not reflected in the prediction intervals.** The intervals shown by `forecast()` only propagate the ARIMA-error innovation variance of the Victoria model. They do **not** account for:

- **Uncertainty in the CPI forecast.** We plugged in the point forecasts $\widehat{\text{CPI}}$ as known covariates, so the variability coming from the CPI ARIMA model (and from re-inflating by a multiplicative, noisy CPI) is ignored.
- **Parameter estimation uncertainty** of the Victoria dynamic regression (coefficients on dummies, trend and the ARIMA parameters are treated as known at their point estimates).
- **Model uncertainty / specification risk**: we assumed one knot at 2008 Q1, a piecewise-linear trend, four deterministic seasonal dummies, and an ARIMA(1,0,0)(0,0,1)$_4$ error process. Any of these choices could be wrong, and the intervals do not widen to reflect that.

5. State space models

Consider

$$T_t = T_{t-1} + \epsilon_t, \quad S_t = 0.8\, S_{t-4} + \eta_t, \quad Y_t = T_t + S_t + V_t,$$

with $\epsilon_t \sim WN(0, \sigma_1^2), \eta_t \sim WN(0, \sigma_2^2), V_t \sim WN(0, \sigma_3^2)$, mutually uncorrelated.

a. **Linear Gaussian state space form.**

Because S_t depends on S_{t-4}, we must keep three extra lags in the state. Let

$$\mathbf{X}_t = \left(T_t,\ S_t,\ S_{t-1},\ S_{t-2},\ S_{t-3}\right)^\top \in \mathbb{R}^5.$$

Then

$$\mathbf{X}_t = F\,\mathbf{X}_{t-1} + \mathbf{w}_t, \qquad Y_t = H\,\mathbf{X}_t + V_t,$$

with

$$F = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{w}_t = \begin{pmatrix} \epsilon_t \\ \eta_t \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad Q = \mathrm{Cov}(\mathbf{w}_t) = \mathrm{diag}(\sigma_1^2,\ \sigma_2^2,\ 0,\ 0,\ 0),$$

and

$$H = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad R = \mathrm{Var}(V_t) = \sigma_3^2.$$

Verification: row 1 of $F\mathbf{X}_{t-1} + \mathbf{w}_t$ gives $T_{t-1} + \epsilon_t = T_t$; row 2 gives $0.8\, S_{t-4} + \eta_t = S_t$ (using that the 5-th entry of $\mathbf{X}_{t-1}$ is S_{t-4}); rows 3–5 simply shift $S_{t-1}, S_{t-2}, S_{t-3}$ down one lag. And $H\mathbf{X}_t = T_t + S_t$, so $H\mathbf{X}_t + V_t = Y_t$.

b. **Does** $\lim_{t\to\infty} \mathrm{Var}(S_t - \hat{S}_{t|n})$ **exist?**

Interpreting $\hat{S}_{t|n}$ as the optimal (MSE-minimising) predictor given $Y_{1:n}$ with n fixed and $t \to \infty$, we are asking about long-horizon forecast error for the seasonal component. Since $|0.8| < 1$, the process $S_t = 0.8S_{t-4} + \eta_t$ is a causal stationary AR process (an AR(1) in the lag-4 operator) with

$$\gamma_S(0) = \mathrm{Var}(S_t) = \frac{\sigma_2^2}{1 - 0.8^2} = \frac{\sigma_2^2}{0.36}.$$

As the forecast horizon $h = t - n \to \infty$, $\hat{S}_{t|n} \to E[S_t] = 0$ and the forecast error variance approaches the unconditional variance. So the limit **does exist** and

$$\lim_{t\to\infty} \mathrm{Var}(S_t - \hat{S}_{t|n}) = \gamma_S(0) = \frac{\sigma_2^2}{1 - 0.8^2} = \frac{\sigma_2^2}{0.36} = \frac{25\,\sigma_2^2}{9}.$$

c. **Does** $\lim_{t\to\infty} \mathrm{Var}(T_t - \hat{T}_{t|n})$ **exist?**

T_t is a random walk, so $T_t = T_n + \sum_{k=n+1}^{t} \epsilon_k$ for $t > n$. The best predictor given $Y_{1:n}$ is $\hat{T}_{t|n} = \hat{T}_{n|n}$ (a random walk's forecast is flat), so

$$\mathrm{Var}(T_t - \hat{T}_{t|n}) = \mathrm{Var}(T_n - \hat{T}_{n|n}) + (t - n)\,\sigma_1^2.$$

As $t \to \infty$ the second term diverges linearly, so the limit **does not exist** (it is $+\infty$). This reflects the familiar fact that long-run forecasts of a random walk have unbounded uncertainty, while the stationary seasonal component's forecast uncertainty is bounded by its marginal variance.